DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

PROGRAMME:ELECTRONICS AND COMPUTER ENGINEERING

Kerala Legislative Assembly Digital Document Assistance System

ABSTRACT

Kerala Legislative Assembly Digital Document Analysis System aims to democratize access to public legislative records by coupling open-source Retrieval-Augmented Generation (RAG) architecture with locally hosted large language models (LLMs). Existing document repositories provide only rudimentary keyword search and require expertise in legal terminology, creating a barrier for citizens, researchers, and policymakers. Our solution transforms this landscape by enabling natural-language querying in both Malayalam and English, returning concise, context-rich answers supported by the original source text.Technically, the platform integrates four core layers: (1) high-accuracy OCR for scanned proceedings and bills, (2) vectorized document embeddings stored in an efficient similarity index (e.g., FAISS or Chroma), (3) lightweight LLMs orchestrated through Ollama for on-premises inference, and (4) a user-friendly web interface with role-based access control. This fully self-hosted stack safeguards data sovereignty, minimizes recurring costs, and ensures offline operability within government infrastructure.In sum, the system offers a scalable, privacy-preserving pathway to make parliamentary knowledge truly accessible, setting a new benchmark for open governance in India.

Keywords**:** *Retrieval-Augmented Generation (RAG),Large Language Models (LLMs), Optical Character Recognition (OCR).*

Reference:

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